



10132 - Ultradeep H α narrow-band imaging of a $z=1.85$ massive galaxy group: RAM-pressure stripping, satellite mass function and clump formation rate

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
HalpHa_group_185				
	1		NIRCam Imaging	(1) GROUP-185

ABSTRACT

We propose NIRCam narrow-band (NB) F187N HalpHa imaging of a spectroscopically-confirmed massive galaxy group at $z=1.85$ in the EGS field ($M_H \sim 4 \times 10^{13} M_{\odot}$). The group contains >10 already confirmed massive ($M_{\text{star}} > 10^{10} M_{\odot}$) galaxies, a star-bursting Brightest Group Galaxy (BGG) in the process of assembling, and significant diffuse intra-halo light that has already been detected in several NIRCam broad-band filters.

By a stroke of good fortune, the HalpHa line emission of the group is redshifted perfectly into the center of the F187N narrow band filter. We will create a high-spatial-resolution panoramic HalpHa map of the group by combining our proposed narrow band observations with existing JWST deep broad-band images. Based on TNG100 we expect 100 star-forming galaxy detections down to $M_{\text{star}} \sim 10^8 M_{\odot}$ within the virial radius, and possibly more in the larger-scale structure.

With just 4.8 hours of JWST time, including overheads, we will use the HalpHa emission line image to:

- (i) measure the shape of the star-forming galaxy mass-function down to $M_{\text{star}} \sim 10^8 M_{\odot}$ in a group (i.e., single halo) at $z=1.85$, which places first constraints on the Missing Satellite Problem at high- z and to test alternative dark-matter models predictions;
- (ii) spatially resolve the star-formation in high- z group galaxies by detecting all massive clumps, including the youngest, and to thus measure the clump formation rate;
- and (iii) obtain the highest-redshift probe of RAM-pressure stripping (RPS) yet, providing first observational clues of its relevance for quenching in the earlier Universe.

OBSERVING DESCRIPTION

Simultaneous narrow-band F187N and medium-band F360M imaging observations in the EGS field, using JWST/NIRCam. There is a single target, a galaxy group at $z=1.85$, observed for a total time of 4.8 hours (3 hours on source), with a FULLBOX, 6TIGHT dither pattern to close the gap between modules. A special requirement in phase 2 is that an offset may be applied so that our target position is near the center of module A. Additionally, we constrain the position angle to be between 90 and 150 degrees, in order to match our sky coverage closely with that of the publicly-available CEERS data. We use a large dither pattern in order to fill the intra-module gaps, to obtain a good overall coverage and therefore enable accurate sky subtraction, as necessary for our science goals.

Proposal 10132 - Targets - Ultradeep Halpna narrow-band imaging of a $z=1.85$ massive galaxy group: RAM-pressure stripping, satellit...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GROUP-185	RA: 14 19 0.2100 (214.7508750d) Dec: +52 49 47.80 (52.82994d) Equinox: J2000		
Comments: Category=Clusters of Galaxies Description=[Galaxy groups]					

Proposal 10132 - Observation 1 - Ultradeep Halpha narrow-band imaging of a z=1.85 massive galaxy group: RAM-pressure stripping, ...

Observation	Proposal 10132, Observation 1										Tue Mar 31 16:01:27 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRCam Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	GROUP-185	RA: 14 19 0.2100 (214.7508750d)								
			Dec: +52 49 47.80 (52.82994d)								
			Equinox: J2000								
Template	Comments: Category=Clusters of Galaxies Description=[Galaxy groups]										
	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		6TIGHT		SMALL-GRID-DITHER				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F360M	MEDIUM2	9	1	12	12	10564.982		
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 90 to 150 Degrees (V3 89.94737309 to 149.94737309) Visits Same PA Fiducial Point Override NRCBS_FULL										